Operational spin states, reversible interaction and finite closure

For classical unit orientations with all Borel preparations and restricted affine product tests, local rotations close on a finite-dimensional ball state space. The complete reversible interaction $H = Ju_z v_z$, however, does not descend to its moment tensor, and no finite enlargement by bounded observables repairs exact descent on any nonzero time interval. These are explicit classical countermodels and proofs. They identify the missing operational closure premise in source-backed reconstructions of quantum composition; they do not select a positive universal action parameter.

The model below keeps its preparations, effects and transformations explicit. It incorporates C125 and C127–C129 and the associated source-premise map. Historical coordinator/model checks are internal. The reconstruction theorems are cited with their separate hypotheses and reading levels; their complete proofs have not been independently audited here.

A ball of local states and a minimal composite

Let the hidden state be $u \in S^2$, with every Borel probability measure μ admitted as a preparation. A local measurement is a finite family

$$e_i(u) = a_i + b_i \cdot u, \qquad a_i \geq |b_i|, \qquad \sum_i a_i = 1, \quad \sum_i b_i = 0.$$

Its probabilities depend exactly on $r = \int u \, d\mu$. Every point of the closed unit ball is achieved by mixing opposite orientations, and the tests $(1 \pm n \cdot u)/2$ determine r. Thus operational equivalence is equality of r, pure states are the unit sphere, and the unnormalized state cone has linear dimension four. If a measurement distinguishes k states perfectly, then each successful outcome obeys $1 = a_i + b_i \cdot r_i \leq 2a_i$, hence $k \leq 2$. Opposite poles attain capacity two. Local $SO(3)$ rotations descend exactly to $r \mapsto Rr$ and connect all pure states continuously.

For n components, admit all joint Borel measures on $(S^2)^n$, product local tests, mixtures and coarse-grainings, local rotations and permutations. Put $v(u) = (1, u)$. All test probabilities depend on

$$z = \int v(u_1) \otimes \cdots \otimes v(u_n) \, d\mu.$$

Product tests span these coordinates. The composite is exactly the convex hull of product ball states, with linear dimension 4^n. Each product effect is at most 2^n times its value at the product of zero vectors. This survives mixing and coarse-graining; a test distinguishing k states must spend at least 2^{-n} of its normalization per successful outcome, so $k \leq 2^n$. Product pole tests attain 2^n.

These are stipulated terminal measurements, not unrestricted hidden-state readouts or invasive instruments. If every hidden orientation event were measurable, the operational state would distinguish the full probability measures. The model is valid with its local transformations; additional interactions must be tested for descent.

What reconstruction premises add

This ball model satisfies local pure-state connectivity but not the whole axiom package of either reconstruction below:

Source and selected reading	Additional premises	Classical comparison
Hardy, version 4, §§1, 4, 7; B67, passages	Probability, simplicity, subspace and composition axioms, plus continuous reversible pure-state connectivity	A finite simplex has isolated vertices and only vertex permutations; the orientation ball evades this particular failure
Chiribella–D'Ariano–Perinotti, version 3, §II and §§III.1–III.2; B67, passages	Causality, perfect distinguishability, ideal compression, local distinguishability, pure conditioning and purification with essential uniqueness	Pure states of the minimal classical composite have pure marginals, so a mixed local state has no purification

The purification assertion follows directly. The product-pure set is compact; every extreme point of its convex hull is product-pure by a finite convex representation. Conversely a unit-vector marginal forces that orientation almost surely, successively in every factor, so product-pure points are extreme. Every marginal of a pure composite is therefore pure. The state $r = 0$ has no pure extension by any number of these components.

Hardy's subspace condition also fails explicitly. In two components, coarse-grain the local pole tests along n to equal signs:

$$e_{\mathrm{same}}(u, v) = \tfrac{1}{2}[1 + (n \cdot u)(n \cdot v)].$$

Probability one requires $(n\cdot u)(n\cdot v) = 1$ almost surely, leaving only (n, n) and $(-n, -n)$. The corresponding capacity-two face is a segment, whereas an elementary capacity-two system here is a three-dimensional ball. Local continuity and multiplicative capacity/dimension therefore do not select quantum composition.

Hardy's moving-ball comparison motivates closure throughout a transformation: coarse-graining intermediate positions into a bit does not make their reduced evolution reversible. Likewise a definite mechanical microstate is not by itself a CDP operational purification. The full reconstruction packages remain conditional inputs. Originals: Hardy and CDP. Only the indicated passages and hypothesis comparisons were read; no claim is made about all continuous classical phase spaces or the infinite-dimensional extensions.

A complete classical interaction fails moment descent

Fix spin magnitudes $S_A, S_B > 0$ with action units and $J > 0$ with energy units. On $S^2 \times S^2$ use

$$\{u_i, u_j\} = S_A^{-1}\epsilon_{ijk}u_k, \qquad \{v_i, v_j\} = S_B^{-1}\epsilon_{ijk}v_k, \qquad \{u_i, v_j\} = 0, \qquad H = Ju_zv_z.$$

With $\dot f = \{f, H\}$ and $\Omega_A = J/S_A$, $\Omega_B = J/S_B$,

$$\dot u_x = -\Omega_A u_y v_z, \quad \dot u_y = \Omega_A u_x v_z, \quad \dot u_z = 0,$$
$$\dot v_x = -\Omega_B v_y u_z, \quad \dot v_y = \Omega_B v_x u_z, \quad \dot v_z = 0.$$

Norms and energy are conserved. Smoothness on compact phase space makes the flow complete and reversible, with inverse at negative time. Its frequencies depend on the other spin, so this is an actual hidden interaction.

Prepare the exact pair

$$\mu_A = \delta_{e_y}(du)\tfrac{1}{2}(\delta_{e_z} + \delta_{-e_z})(dv), \qquad \mu_B = \delta_{e_y}(du)\tfrac{1}{2}(\delta_{e_x} + \delta_{-e_x})(dv).$$

Both have $\mathbb{E}[u] = e_y$, $\mathbb{E}[v] = 0$, $\mathbb{E}[uv^T] = 0$, and microscopic energy identically zero. They agree on all original terminal tests. Since $u_z = 0$, v stays constant. In the first preparation,

$$u(t) = (-\operatorname{sign}(v_z) \sin \Omega_A t, \cos \Omega_A t, 0), \qquad \mathbb{E}_{\mu_A}[u_x(t) v_z(t)] = -\sin \Omega_A t.$$

The second retains $u = e_y$ and has zero cross expectation. The allowed product effect $(1 + u_x)(1 + v_z)/4$ consequently gives

$$p_A(t) = \tfrac{1}{4}(1 - \sin \Omega_A t), \qquad p_B(t) = \tfrac{1}{4}.$$

These differ at every sufficiently small positive time. No single-valued update, linear or nonlinear, on the initial moment tensor can represent this flow for all preparations. Both evolved measures still give valid minimal-composite states: the failure is descent to the quotient. Its missing moment is exposed by

$$\frac{d}{dt}\mathbb{E}[u_x v_z] = -\Omega_A \mathbb{E}[u_y v_z^2].$$

Source/proof context: B72, written spin conventions and internal comparison. The supplied S_A, S_B are not derived Planck constants.

No finite observable enlargement repairs this interaction

Let V be a finite-dimensional real space of bounded Borel functions on $X = S^2 \times S^2$, compared pointwise, containing 1 and the original terminal effects. Exact closure would require $U_t V \subset V$, where $U_t f = f \circ \Phi_t$, for every t in a nonzero interval. A forward interval is sufficient for the contradiction. The excluded case $J = 0$ has static finite closure.

In the complexification, $w = u_x + i u_y \in V_{\mathbb{C}}$ obeys exactly

$$U_t w = w \exp(i \Omega_A t v_z).$$

For distinct $t_1, \ldots, t_m$, a relation $\sum_j c_j U_{t_j} w = 0$ restricted to $u = e_x$ and $v = (\sqrt{1 - z^2}, 0, z)$, $-1 < z < 1$, gives

$$\sum_j c_j \exp(i \Omega_A t_j z) = 0.$$

Derivatives at $z = 0$ of orders $0, \ldots, m - 1$ form a Vandermonde system with determinant $\prod_{j<l}(i\Omega_A t_l - i\Omega_A t_j) \neq 0$. Every coefficient vanishes. Arbitrarily many time translates are therefore independent, contradicting finite dimension. No regularity of added observables is needed. The smooth infinitesimal hierarchy is equivalently $L^n w = (i\Omega_A)^n w v_z^n$, but the exact time-orbit proof establishes closure failure without a truncation argument.

Nor can a nonlinear update of finitely many expectations evade it. If the expectation of g is determined by those of $f_0 = 1, f_1, \ldots, f_d$ for **every** probability measure, then g lies in their real span. Otherwise finite evaluation linear algebra supplies points x_j and weights a_j with $\sum_j a_j f_k(x_j) = 0$ for every k, but $\sum_j a_j g(x_j) \neq 0$. If no such relation existed, $g(x)$ would define a linear functional on the span of the evaluation vectors and hence be a combination of the f_k. Since $\sum_j a_j = 0$, positive and negative weights have equal nonzero mass; normalizing them gives two admitted atomic preparations with equal retained expectations and unequal g expectation. Apply this to $g = U_t f_k$.

Thus exact finite expectation coordinates require an invariant finite observable span, already excluded above. Arbitrary encodings of whole measures without affine probabilistic mixing are outside this theorem. Infinite observable dimension does not by itself imply infinite perfect-distinguishability capacity or quantum structure. Restricted preparations and finite accuracy change its hypotheses. Source/proof context: B74, written proof and literature boundary.

Reversible operational interaction: the source theorem

A different premise demands a connected reversible **linear** group on the fixed operational tensor coordinates, containing local rotations and a genuine interaction. Under $\rho(r) = (I + r \cdot \sigma)/2$, $E = aI + b\cdot\sigma$, local probabilities are $\mathrm{tr}(E\rho) = a + b\cdot r$. The tensor coordinates become Hermitian matrices and the minimal composite becomes exactly separable density matrices. No entangled state was inserted by this coordinate identification.

Theorem 1 of de la Torre–Masanes–Short–Müller assumes valid probabilities for every product preparation, group element and product test, with the group containing local unitaries. A larger connected group contains, on a pair with fixed product spectators, an entangling unitary action or its partial-transpose conjugate. The first leaves the minimal composite by creating entanglement. For the second, choose a product input whose partially transposed input is entangled by that unitary: the output is the partial transpose of a pure entangled state. Its Schmidt eigenvalues include $\pm s_1 s_2$, $s_1 s_2 > 0$, so it cannot be separable positive semidefinite. Either branch excludes an interacting extension preserving the minimal cone.

Theorem 2 additionally uses interchangeable identical systems and three-system consistency to eliminate the partial-transpose alternative. Entangling-gate universality and ancilla preparation, joint evolution, measurement and discard closure supply the full quantum preparation, measurement and completely positive channel conclusion. Those assumptions must accompany that conclusion. The theorem concerns finite numbers of identical qubits, not arbitrary local state spaces.

The source statements, assumptions and these consequences were read at passage level in B70. The technical Lie-algebra and gate-universality proofs are borrowed results, not independently audited proofs of this repository. The exact hidden Hamiltonian above does not contradict the theorem: it lacks reversible operational descent, which no finite moment repair supplies under its preparation hypotheses.

The checked first-order generator restriction

A specific source step does have a full written verification here (C129). Let X be a real $4^n \times 4^n$ matrix with e^{tX} admissibly reversible for a two-sided neighbourhood of zero. Independently available unit Bloch vectors in each input and effect give

$$p_{b,a}(t) = 2^{-n} v(b)^T e^{tX} v(a) \in [0,1].$$

The expansion is

$$p(t) = p(0) + 2^{-n} t\, v(b)^T X v(a) + 2^{-n} \tfrac{t^2}{2} v(b)^T X^2 v(a) + O(t^3).$$

An orthogonal pair $b_1 = -a_1$ gives $p(0) = 0$, so the two-sided minimum requires $v(-a_1, b_2, \ldots)^T X v(a) = 0$ and the corresponding X^2 quadratic form nonnegative. With $b_k = a_k$ at every site, $p(0) = 1$ and $v(a)^T X^2 v(a) \le 0$. These necessary derivative signs are not sufficient finite-time positivity conditions. A forward semigroup alone permits a nonzero first derivative at a boundary.

Fix spectator input and output indices. The vectors $v(e_i) \pm v(-e_i)$ span $\mathbb{R}^4$, so independent product spanning implies the local 4×4 slice M satisfies $v(-a)^T M v(a) = 0$ for every $|a| = 1$. Writing $M = \left(\begin{smallmatrix} c & r^T \\ s & D \end{smallmatrix}\right)$, this says $c + (r - s)\cdot a - a^T D a = 0$. Values at $a, -a$ give $r = s$; at e_i they give $D_{ii} = c$; at $(e_i + e_j)/\sqrt{2}$ they give $D_{ij} + D_{ji} = 0$. Thus exactly

$$M = cI_4 + \begin{pmatrix} 0 & r^T \\ r & K \end{pmatrix}, \qquad K^T = -K.$$

These slices form the seven-dimensional space $\mathcal{L} = \mathcal{A} \oplus \mathcal{B} \oplus \mathcal{I}$, with three antisymmetric spatial directions, three symmetric time–space directions and one scalar. At each site X lies

in the tensor space with that site restricted to $\mathcal{L}$. Expand in a basis adapted to $\mathcal{L}$ and a complement; intersecting all site restrictions leaves

$$X \in \mathcal{L}^{\otimes n}.$$

This is a necessary ambient linear space, not the Lie algebra of an admissible group. Normalization, the second-order inequalities and later source steps remain. Neither local rotation averaging nor ancilla/discard closure was needed for this first-order restriction. Source: the held original, PDF pp. 3–4, equations (4), (6), (8)–(14); B77 derivation.

Consequence for STATE

Finite operational capacity and smooth classical reversibility coexist with failure of composite descent. Exact closure for the specified interacting spin model requires infinite observable dimension or a changed preparation, effect or dynamics class. Quantum reconstruction needs its stated additional composition premises. All probability coordinates above are dimensionless; matching their transformations to mechanical time, energy and phase remains separate from determining quantum structure or a universal physical $\hbar$.